

IoMT model creation

This document presents a step-by-step tutorial on how to create an IoMT model using the IoMT profiles.

How-to objective

Creation of an IoMT UML model with a set of classes.

This how-to is separated into 2 major steps:

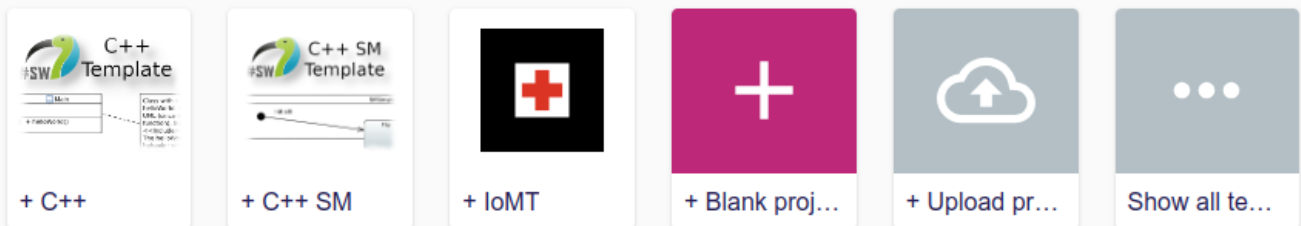
1. **Model creation:** how to create IoMT UML models and diagrams.
2. **Model edition:** how to edit the model elements.

The URL of PapyrusWeb is : <http://localhost:8080>

Model Creation

PapyrusWeb enables the creation of different kinds of projects. The start page provides a set of templates, or an existing project may be reopened.

Create a new project



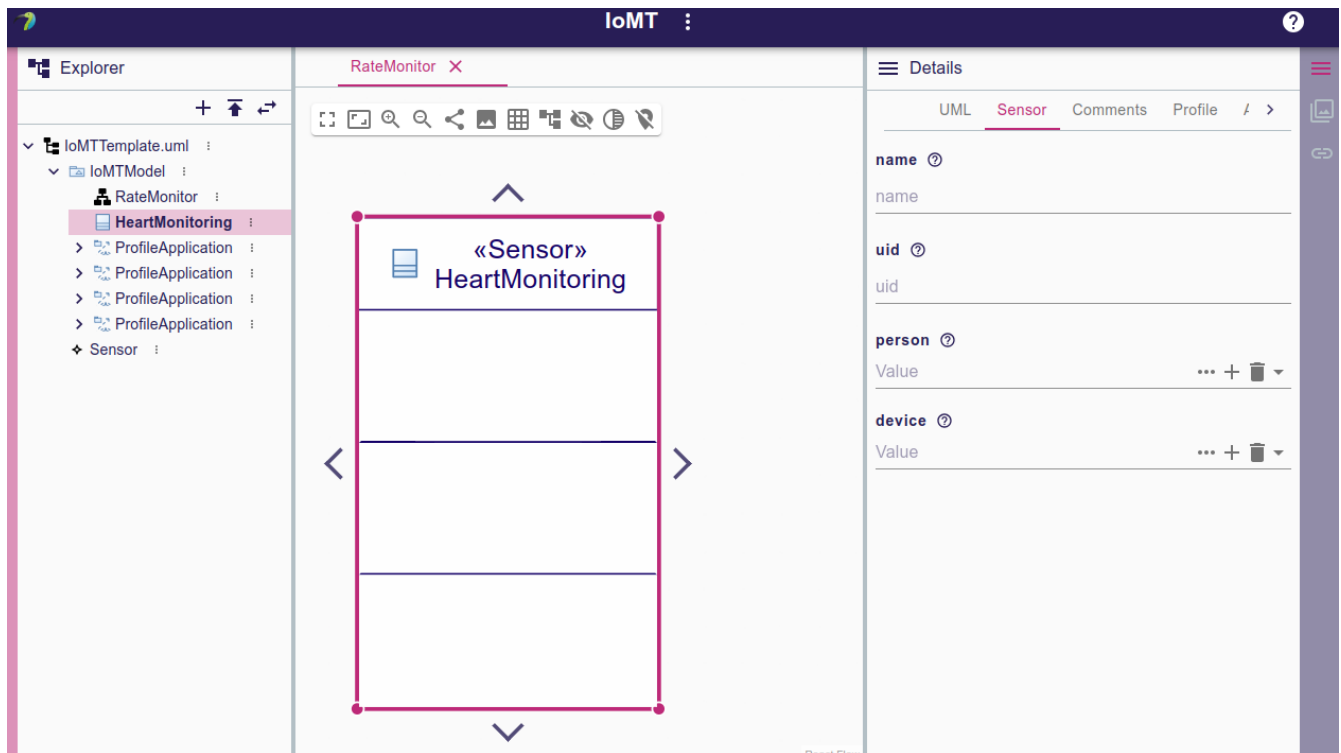
Existing Projects

No projects found, start by creating one

We select the **IoMT** template. It performs the following actions:

- **Papyrus Software Designer**
 - creates a IoMT UML project template with one Class named *HeartMonitoring*. This class has the stereotype *Sensor* applied.
 - applies 4 profiles with IoMT concepts: - *IoMTAssets*, *IoMTProfile*, *IoMTComponentCategories* and *Assertions*. Not all concepts need to be used, only the ones necessary for each specific architecture definition.

The project should have a structure as illustrated below:



A project has three main components:

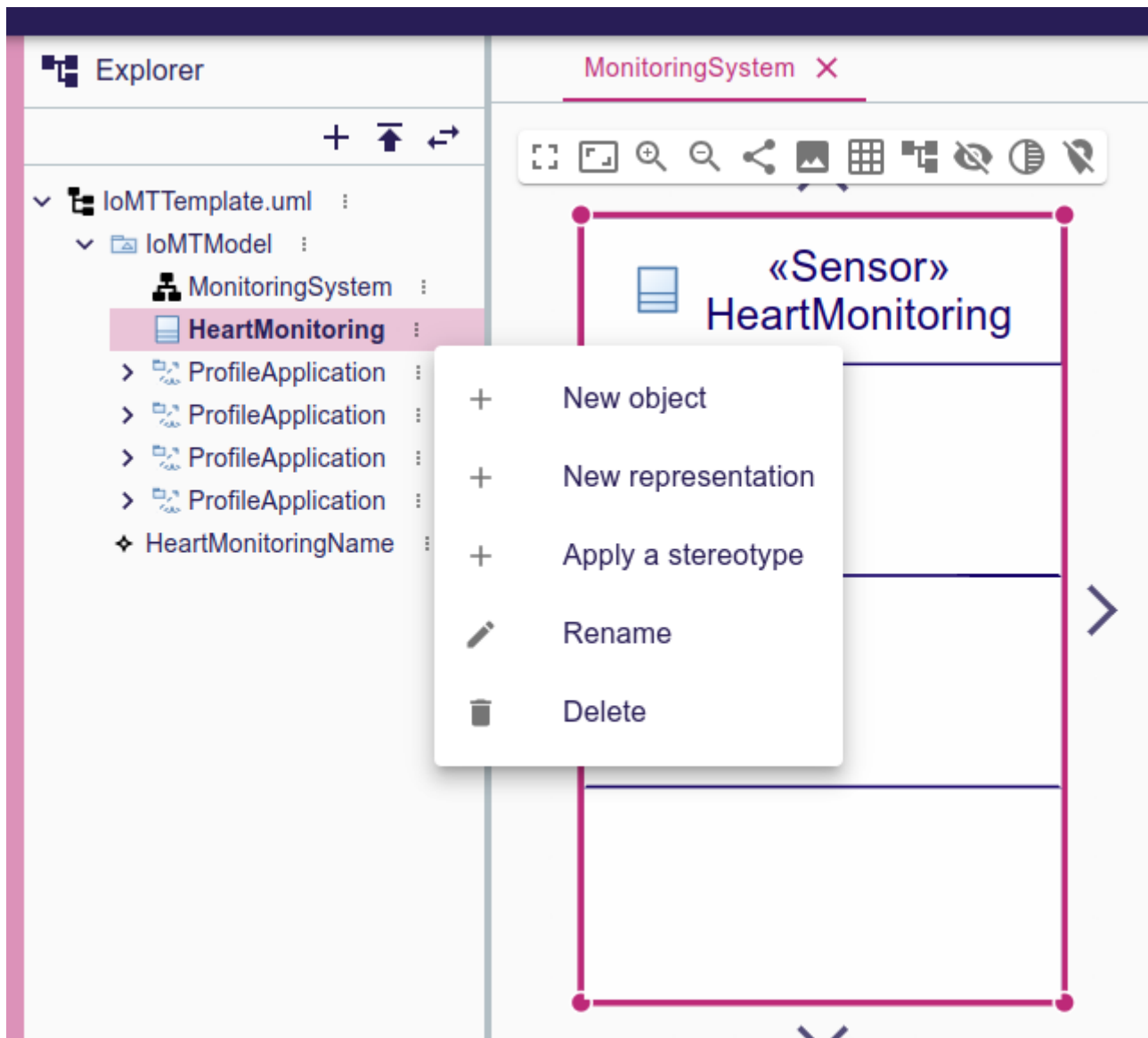
- **Model explorer** (right side): the model explorer enables the navigation and edition in the existing elements. Each model element has a contextual menu with a specific set of actions. For instance, it enables creating child elements of a given Class.
- **Diagram editor** (center): the diagram editor enables creating visual representation of the elements of the model
- **Property view** (left side): the property view enables the edition of properties of each selected element of the model. The selected properties are referent to the *Sensor* specific properties.

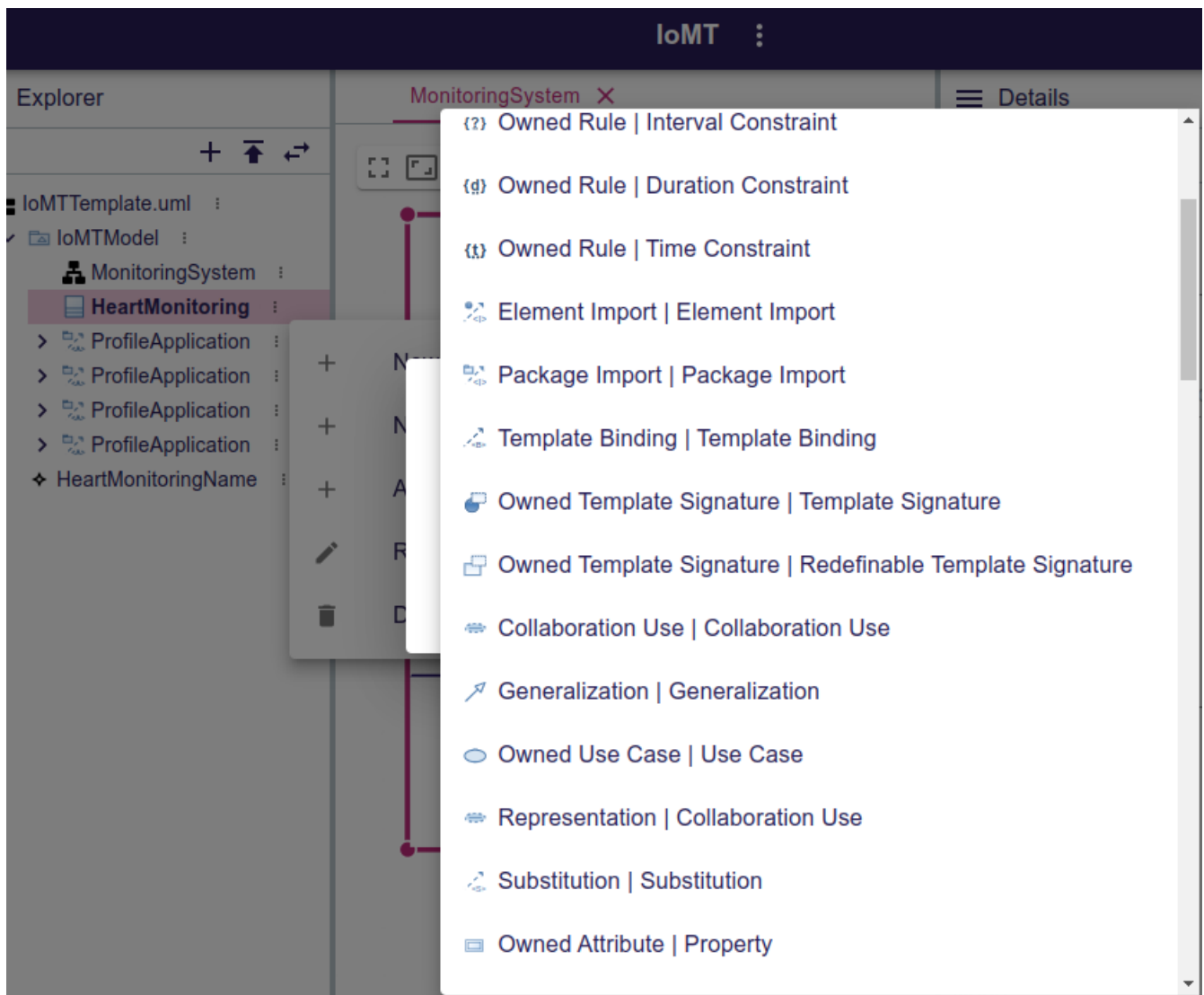
All the possible options are explained in the [PapyrusWeb help documentation](#)

Model edition

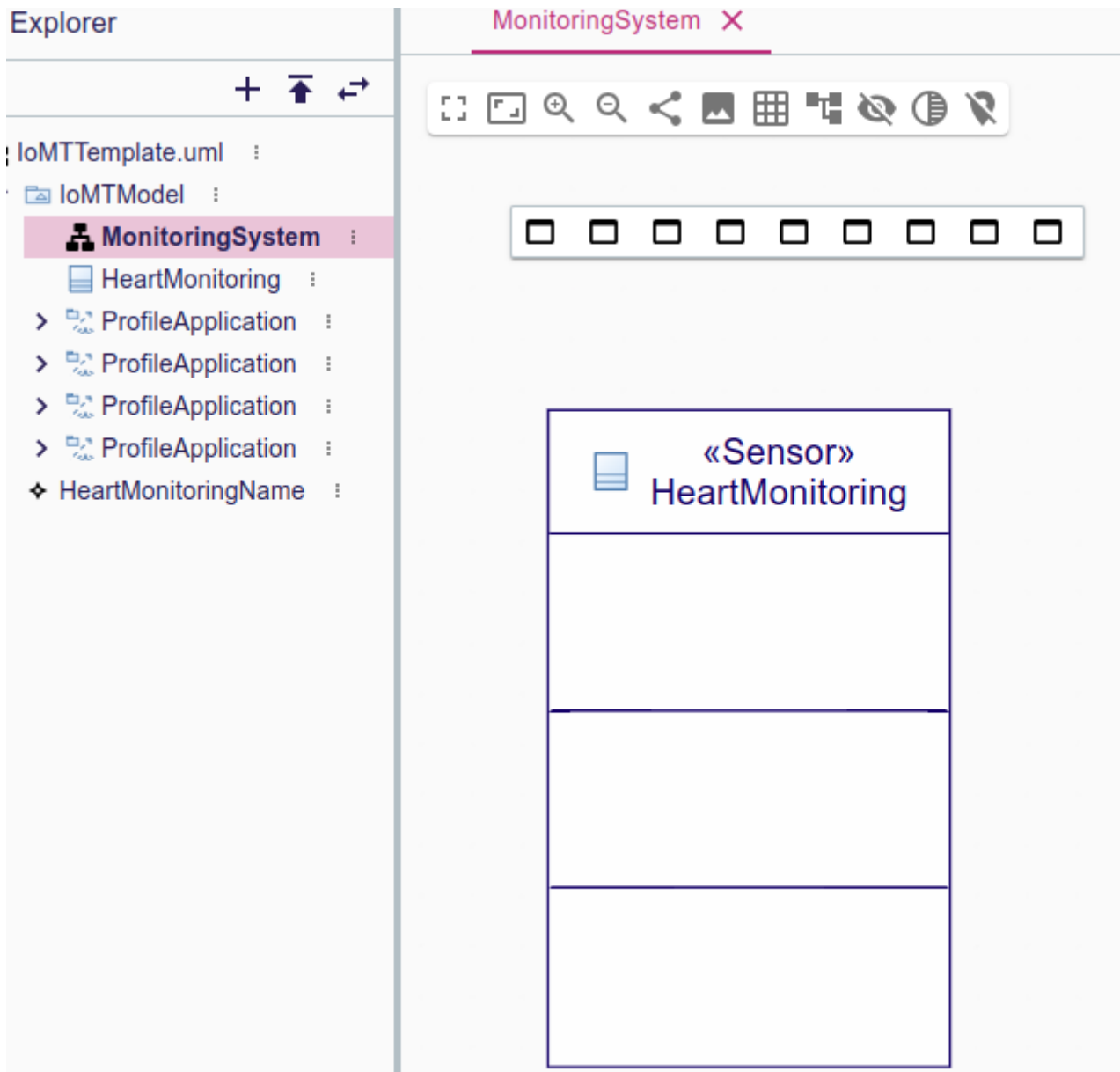
The model may be edited using the diagram editor or the model explorer. Both views are synchronized.

- In the model explorer, the user selects an element and chooses a child element to be created in the dedicated menu.



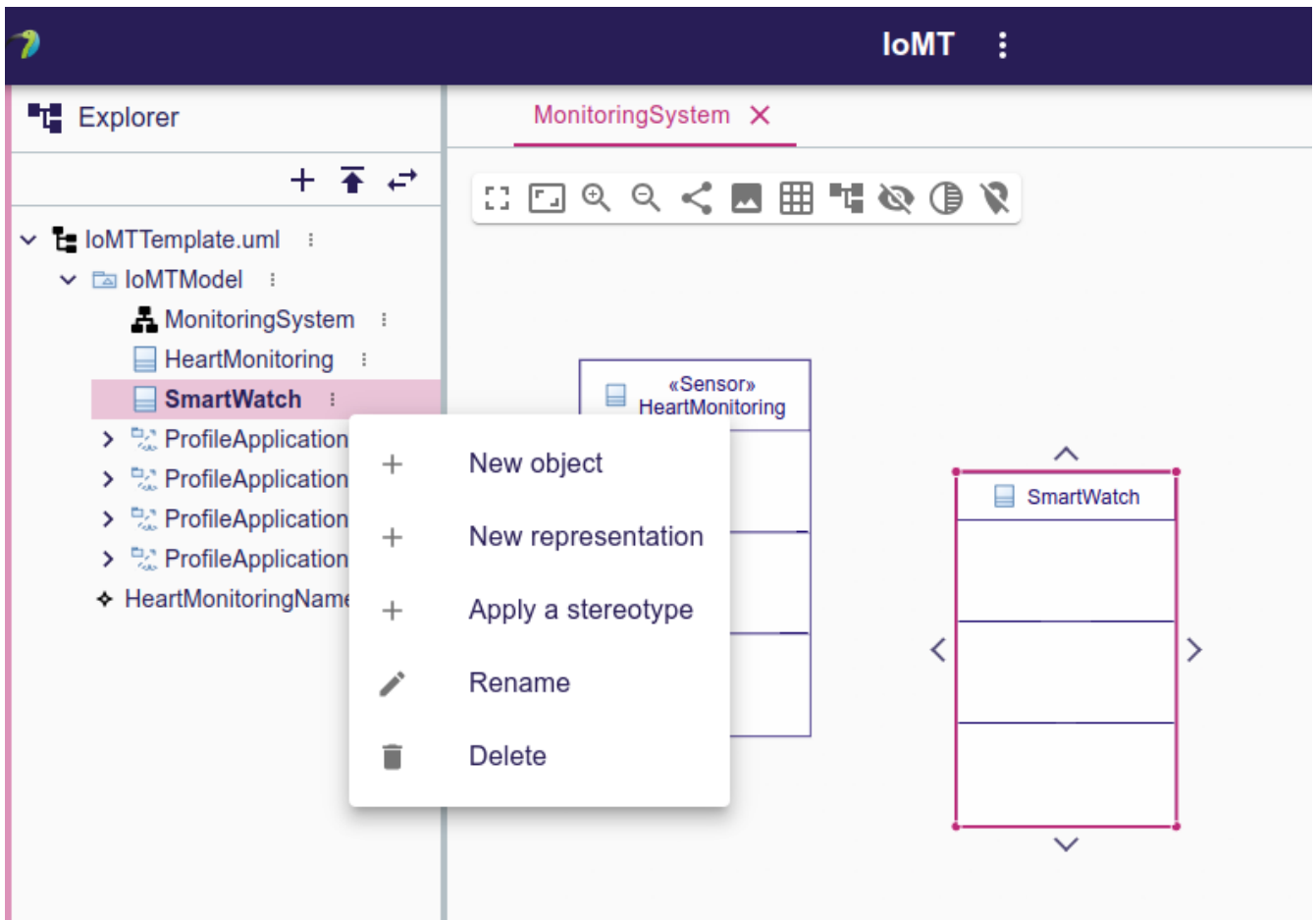


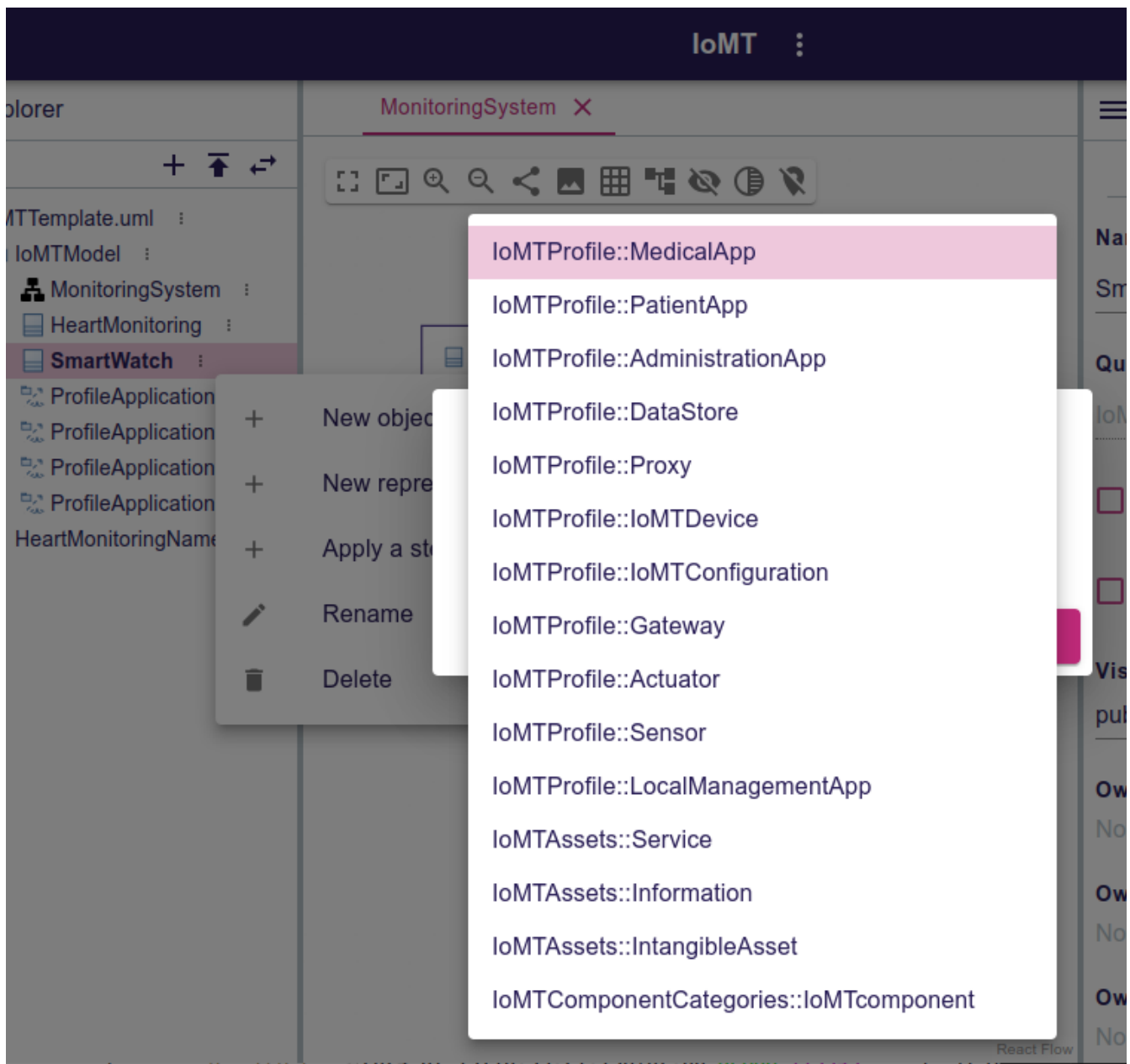
- In the representation view (diagram), the user clicks in the diagram to open the contextual palette and selects the new desired element.



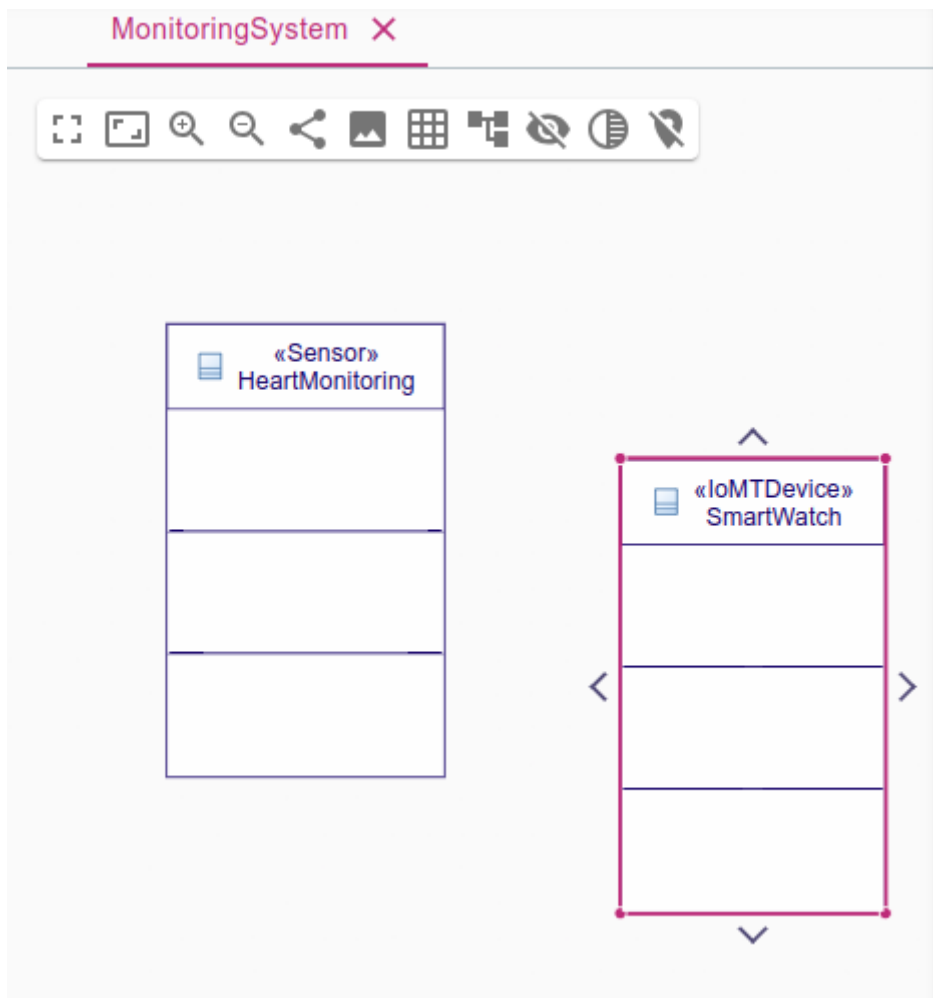
For each created class, the property view show its standard UML properties and the specific IoMT properties. These properties are available once we apply specific profiles for the classes.

Now we create a *SmartWatch* class, which is a *IoMTDevice* relating to the *HeartMonitoring Sensor*. This class is initially a plain UML class. It is necessary to apply a specific stereotype to promote it to an IoMT specific class. We click in the class in the model explorer and we select *Apply Stereotype* to choose among the possible IoMT elements.





We select the `IoMTProfile::IoMTDevice` stereotype. The class is now stereotyped with *IoMTDevice*, as shown below.



The model can be enriched to add any other IoMT specific elements.